

Visitor Management Backend -- Technical Documentation

Spring Boot 4.1.0 · Java 21 · PostgreSQL 17 · Hibernate 7.4

1. Architecture Overview

```

-----
-                               REST Controllers                               -
-   TimeSlotController · VisitorController                                   -
-   RegistrationController · BriefingController                             -
-----
-                               Services                                       -
-   SchedulingServiceImpl · RegistrationServiceImpl                         -
-   SchedulingRules (constants)                                           -
-----
-                               Repositories (JPA)                             -
-   TimeSlotRepository · VisitorRepository                                 -
-   RegistrationRepository · BriefingRepository                           -
-----
-                               JPA Entities                                   -
-   TimeSlot · Visitor · Registration · Briefing                           -
-   (extend BaseEntity: id + createdAt + updatedAt)                       -
-----
-   core/ -- shared infrastructure                                         -
-   BaseEntity · GlobalExceptionHandler · ApiError                       -
-   ResourceNotFoundException · ConflictException                         -
-   BusinessException                                                    -
-----

```

Packages:

- core/ -- shared infra only (BaseEntity, exceptions, handler)
- modules/visitemanagement/ -- Scheduling + Registration sub-modules
- Enums live in the module package (not core/)

2. Database Schema

Entity-Relationship Diagram

```

-----
-   time_slot   -   visitor   -
-----
- id            BIGINT- PK      - id            BIGINT- PK -
- slot_date     DATE  - NOT NULL- full_name    VARCHAR(150) - NOT NULL
- start_time    TIME  - NOT NULL- group_size  INT      - NOT NULL, DEFAULT 1
- end_time      TIME  - NOT NULL- email       VARCHAR(200) - UNIQUE
- max_capacity  INT    - NOT NULL- phone       VARCHAR(30)  -
- status        VARCHAR(20)- NOT NULL- language    VARCHAR(40)  -
- guide_id      BIGINT -          - visitor_type VARCHAR(20) - NOT NULL
- created_at    INSTANT - NOT NULL- special_needs VARCHAR(500)-
- updated_at    INSTANT - NOT NULL- created_at  INSTANT- NOT NULL
-----
-                               - updated_at  INSTANT- NOT NULL
-----
-   1   -   1   -
-   -   -   -
-   *   -   1   - *
-----
-   registration   -
-----

```

```

-----
- id          BIGINT- PK                      -
- visitor_id  BIGINT- FK ? visitor (NOT NULL) -
- time_slot_id BIGINT- FK ? time_slot (NOT NULL) -
- event_id    BIGINT-                        -
- status      VARCHAR(20)- NOT NULL, DEFAULT 'PENDING' -
- created_at   INSTANT- NOT NULL                -
- updated_at   INSTANT- NOT NULL                -
- UNIQUE (visitor_id, time_slot_id)            -
-----

- 1
-
- 1

-----
- briefing    -
-----
- id          BIGINT- PK                      -
- registration_id - FK ? registration        -
-              - (NOT NULL, UNIQUE)         -
- delivered_at INSTANT-                      -
- staff_member VARCHAR(150)                  -
- signature    VARCHAR(500)                  -
- status      VARCHAR(20)- NOT NULL, 'PENDING' -
- created_at   INSTANT- NOT NULL                -
- updated_at   INSTANT- NOT NULL                -
-----

```

Table Summary

Table	Rows (design)	Purpose
`time_slot`	~500/year	Farm tour calendar slots (2/day max)
`visitor`	~2,000	Visitor database (individuals/groups)
`registration`	~4,000	Links visitor ? time slot
`briefing`	~3,000	Safety briefing per confirmed registration

Enums

Enum	Values	Table Column
`TimeSlotStatus`	`AVAILABLE`, `RESERVED`, `FULL`, `CANCELLED`, `COMPLETED`	`time_slot.status`
`RegistrationStatus`	`PENDING`, `CONFIRMED`, `REJECTED`, `CHECKED_IN`, `CANCELLED`	`registration.status`
`BriefingStatus`	`PENDING`, `DONE`	`briefing.status`
`VisitorType`	`INDIVIDUAL`, `GROUP`	`visitor.visitor_type`

3. REST API Endpoints

Base URL: <http://localhost:8080/api/v1>

3.1 Time Slots -- `/api/v1/time-slots`

Method	Path	Description	Status Codes
`GET`	`/time-slots`	List all slots (optional `?date=YYYY-MM-DD`)	200
`GET`	`/time-slots/{id}`	Get slot by ID	200, 404
`POST`	`/time-slots`	Create a new slot	201, 400, 409, 422
`PUT`	`/time-slots/{id}`	Update slot details	200, 404, 422
`DELETE`	`/time-slots/{id}`	Cancel a slot (soft-delete)	204, 404, 422
`POST`	`/time-slots/{id}/assign-guide`	Assign a guide (`?guideId=N`)	200, 404, 422
`GET`	`/time-slots/availability?date=...`	Availability view for a date	200

Create/Update Request Body:

```

{
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "guideId": 5
}

```

```
}
```

TimeSlotResponse:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "booked": 3,
  "status": "RESERVED",
  "guideId": 5,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

AvailabilityResponse:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "remaining": 7,
  "status": "RESERVED",
  "guideId": 5
}
```

3.2 Visitors -- `/api/v1/visitors`

Method	Path	Description	Status Codes
GET	/visitors	List all visitors	200
GET	/visitors/{id}	Get visitor by ID	200, 404
POST	/visitors	Create visitor	201, 400, 409
PUT	/visitors/{id}	Update visitor	200, 404, 409

VisitorRequest:

```
{
  "fullName": "Alice Dupont",
  "groupSize": 2,
  "email": "alice@example.com",
  "phone": "+241 06 12 34 56",
  "language": "French",
  "type": "INDIVIDUAL",
  "specialNeeds": "Wheelchair access required"
}
```

3.3 Registrations -- `/api/v1/registrations`

Method	Path	Description	Status Codes
GET	/registrations	List (filter by `?timeSlotId=` or `?date=`)	200
GET	/registrations/{id}	Get registration by ID	200, 404
POST	/registrations	Create registration (visitor?slot)	201, 400, 409, 422
PATCH	/registrations/{id}/approve	Approve ? triggers briefing creation	200, 404, 422
PATCH	/registrations/{id}/reject	Reject a pending registration	200, 404, 422
PATCH	/registrations/{id}/check-in	Check in a confirmed visitor	200, 404, 422
PATCH	/registrations/{id}/cancel	Cancel a registration	200, 404, 422

RegistrationRequest:

```
{
  "visitorId": 1,
  "timeSlotId": 10,
  "eventId": null
}
```

3.4 Briefings -- `/api/v1`

```
| Method | Path | Description | Status Codes |
| `GET` | `/registrations/{registrationId}/briefing` | Get briefing for registration | 200, 404 |
| `PATCH` | `/registrations/{registrationId}/briefing/deliver` | Mark briefing delivered | 200, 422 |
```

BriefingDeliverRequest:

```
{
  "staffMember": "Guard Camille",
  "signature": "digital-signature-or-name"
}
```

4. Business Rules

Scheduling Rules (`SchedulingRules.java`)

```
| Rule | Value | Source |
| Max visitors per slot | **10** | Confirmed with module owners (Abdoul) |
| Max slots per day | **2** | Mockup: morning + afternoon |
| Closed day | **Sunday** | All module owners |
| Morning slot | 09:00-11:00 | Default |
| Afternoon slot | 14:00-16:00 | Default |
```

Registration Rules

```
| Rule | Enforcement |
| Cannot register on a **CANCELLED** slot | `BusinessRuleException` |
| No duplicate registration (visitor + slot) | `ConflictException` (409) |
| Group size must fit remaining capacity | `BusinessRuleException` |
| Only **PENDING** ? approve/reject | `BusinessRuleException` |
| Only **CONFIRMED** ? check-in | `BusinessRuleException` |
| Cannot cancel after **CHECKED_IN** | `BusinessRuleException` |
| **CONFIRMED** ? auto-creates `Briefing` (PENDING) | Service layer |
| Briefing delivery requires PENDING status | `BusinessRuleException` |
```

Visitor Rules

```
| Rule | Enforcement |
| Duplicate email ? `ConflictException` (409) | On create and update |
| `@NotBlank` on fullName | Bean validation (400) |
| `@Email` on email | Bean validation (400) |
```

5. Status Flow Diagrams

TimeSlot Status Flow

```

      create
    -----?
    -----?
      -
      -
      -
    register  assign  cancel
      -      guide   -
      ?      -      ?
    -----
    - RESERVED - - - CANCELLED-
    -----
    -
    full -
      ?
    -----
    - FULL -?----- (booked >= capacity)
```

```

-----
-
end_time passes
?
-----
- COMPLETED -
-----

```

Registration Status Flow

```

register -----
- PENDING -
-----? -----
-
-----
- - -
approve reject cancel
- - -
? ? ?
-----
-CONFIRMED - - REJECTED - - CANCELLED -
-----
-
check-in
-
?
-----
-CHECKED_IN -
-----
-
cancel blocked
-
(final state)

```

Briefing Status Flow

```

auto -----
- PENDING - ? created when registration CONFIRMED
-----? -----
-
deliver briefing
-
?
-----
- DONE - ? staffMember + signature + deliveredAt
-----

```

6. Error Handling

All errors return a consistent JSON envelope via `GlobalExceptionHandler`:

```

{
  "status": 404,
  "error": "Not Found",
  "message": "Time slot 999 not found",
  "path": "/api/v1/time-slots/999"
}

```

```

| Exception | HTTP Status | When |
| `ResourceNotFoundException` | 404 | Entity not found |
| `ConflictException` | 409 | Duplicate email, duplicate registration, overlapping slot |
| `BusinessRuleException` | 422 | Invalid state transition, capacity exceeded, Sunday creation |
| `MethodArgumentNotValidException` | 400 | Bean validation failures (`@NotBlank`, `@Email`, etc.) |

```

7. Test Coverage Summary

Layer	Annotation	Tests	Framework
Repository	@DataJpaTest + H2	15	Spring Data + AssertJ
Service	@ExtendWith(MockitoExtension.class)	31	Mockito + AssertJ
Controller	@WebMvcTest + MockMvc	32	MockMvc + Mockito
Context	@SpringBootTest	1	Spring Boot
Total		**79**	

Test Files

```
src/test/java/com/infineonbit/sustainablefarm/
--- SustainableFarmApplicationTests.java
--- modules/visitemanagement/
    --- repository/
        --- TimeSlotRepositoryTest.java      (5 tests)
        --- VisitorRepositoryTest.java       (4 tests)
        --- RegistrationRepositoryTest.java   (4 tests)
        --- BriefingRepositoryTest.java      (2 tests)
    --- service/
        --- SchedulingServiceImplTest.java    (13 tests)
        --- RegistrationServiceImplTest.java  (18 tests)
    --- controller/
        --- TimeSlotControllerTest.java      (9 tests)
        --- VisitorControllerTest.java       (7 tests)
        --- RegistrationControllerTest.java   (11 tests)
        --- BriefingControllerTest.java      (5 tests)
```

8. Running the Backend

```
# Start PostgreSQL + Backend
docker compose up -d backend

# Backend available at
http://localhost:8080

# Swagger UI
http://localhost:8080/swagger-ui/index.html

# Actuator health
http://localhost:8080/actuator/health

# Run tests (Docker, no local JDK required)
docker run --rm -v "$PWD":/src -w /src \
-v "$HOME/.m2":/root/.m2 \
maven:3.9-eclipse-temurin-21 ./mvnw test

# Compile only
docker run --rm -v "$PWD":/src -w /src \
-v "$HOME/.m2":/root/.m2 \
maven:3.9-eclipse-temurin-21 ./mvnw compile
```

9. Technology Stack

Component	Version	Notes
Java	21	LTS
Spring Boot	4.1.0	Modular monolith
Spring Data JPA	4.1.0	Derived queries + @Query
Hibernate ORM	7.4.1	ddl-auto=update (no Flyway)
PostgreSQL	17-alpine	Production database
H2	2.x	Test-only in-memory DB
SpringDoc OpenAPI	2.8.5	Swagger UI auto-generated
Lombok	1.18.46	Boilerplate reduction
Jackson	3.1.4	JSON serialization

| Mockito | 5.23.0 | Service + controller tests |
| AssertJ | 3.27.7 | Fluent assertions |

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